

Colon Cancer Survival Prediction

► Using Machine Learning Models

Presented by

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Data Sources

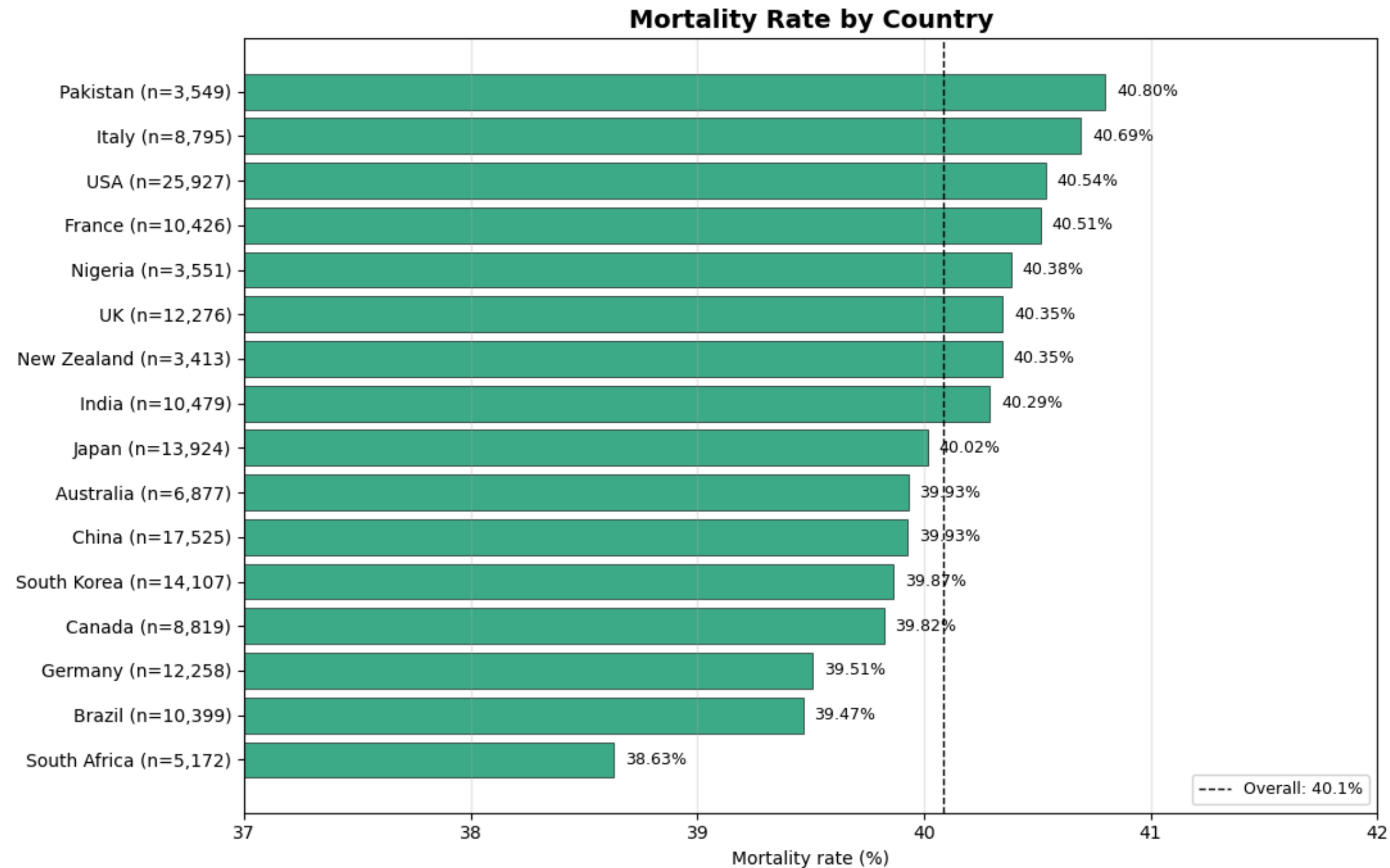
Data Source #	Name/ Short Description	Source URL	Type: - API - Web page - File	List of Fields	Format: - json - xml - csv - sql - other	Have tried to access/collect data with python? yes/no	Estimated data size, number of data points you plan to use
1	NIH Colorectal Cancer Clinical Trials	https://clinicaltrialsapi.cancer.gov/api/v2	API	Healthcare	json	Yes	300 or more
2	Colorectal Cancer Global Dataset	https://www.kaggle.com/datasets/ankushpanday2/colorectal-cancer-global-dataset-and-predictions/discussion	File	Healthcare	CSV	Yes	300 or more
3	Colorectal Cancer Dietary and Lifestyle Dataset – colon cancer risk factors	https://www.kaggle.com/datasets/ziya07/colorectal-cancer-dietary-and-lifestyle-dataset	File	Healthcare	CSV	Yes	300 or more
4	CDC Wonder	https://wonder.cdc.gov/wonder/help/wonder-api.html	Web	Healthcare	XML	Yes	300 or more

Introduction:

- This project explores how machine learning can be used to understand and predict colon cancer survival outcomes across different demographics. Colon cancer is increasingly being diagnosed in younger populations, challenging the long-held assumption that it primarily impacts older adults and making predictive modeling more critical than ever. By leveraging clinical trial intervention data from the NIH Cancer Clinical Trials API, mortality and survival statistics from CDC WONDER, a global colorectal cancer incidence dataset, and lifestyle risk factor dataset. This project aims to build models capable of predicting survival rates across age groups, genders, and countries and external factors. The goal is to demonstrate how combining clinical, epidemiological, and lifestyle data can uncover patterns that support earlier, more targeted interventions in colon cancer care.



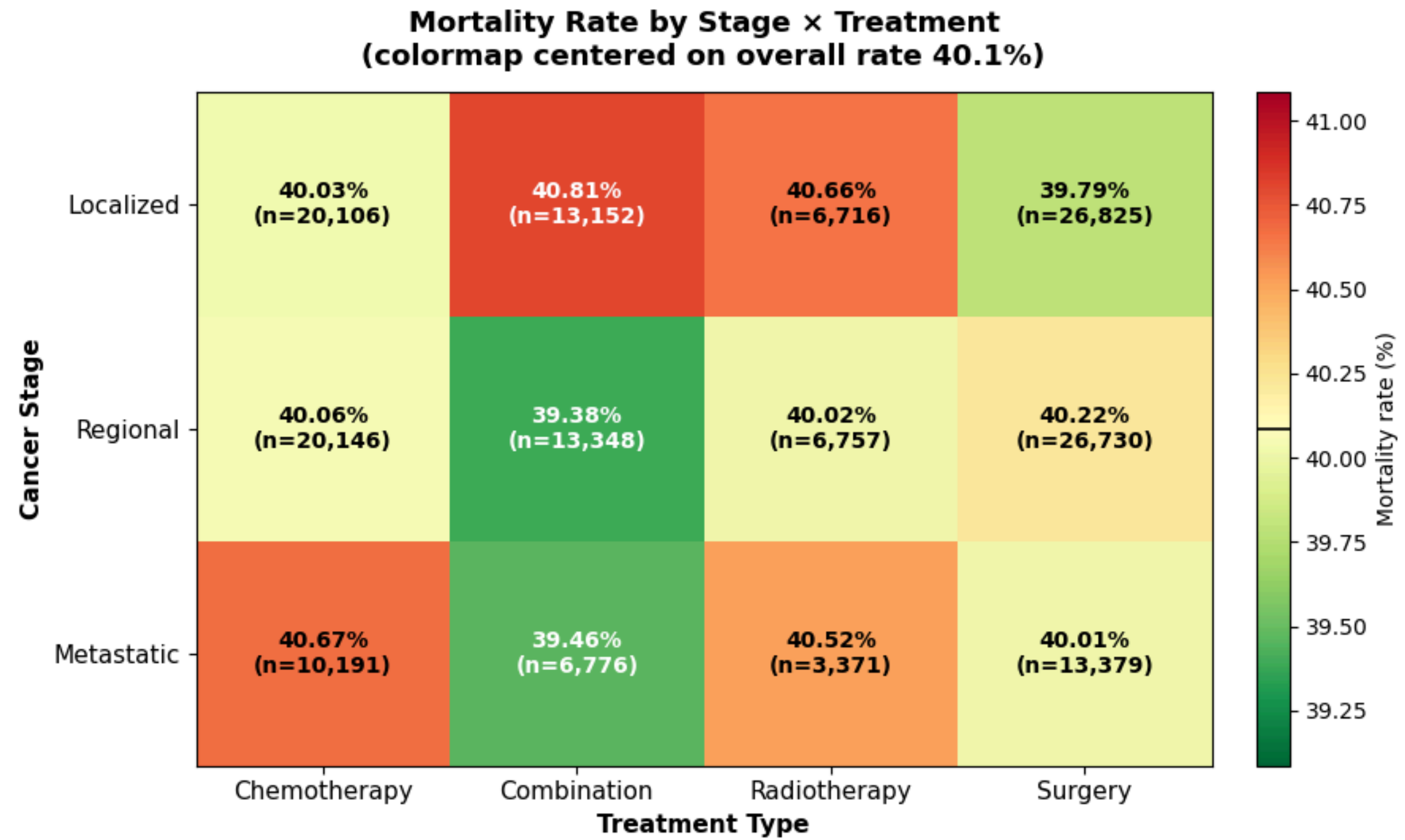
Model Results: Global Colon Cancer Mortality Rates



- Mortality rates across the 16 countries cluster tightly around the overall average of 40.1%, with all values falling within a narrow ~2 percentage point range (38.63%–40.80%). Pakistan, Italy, and the USA sit at the high end, while South Africa, Brazil, and Germany show the lowest rates, though the differences are small and likely reflect minor variation rather than meaningful disparities in outcomes.

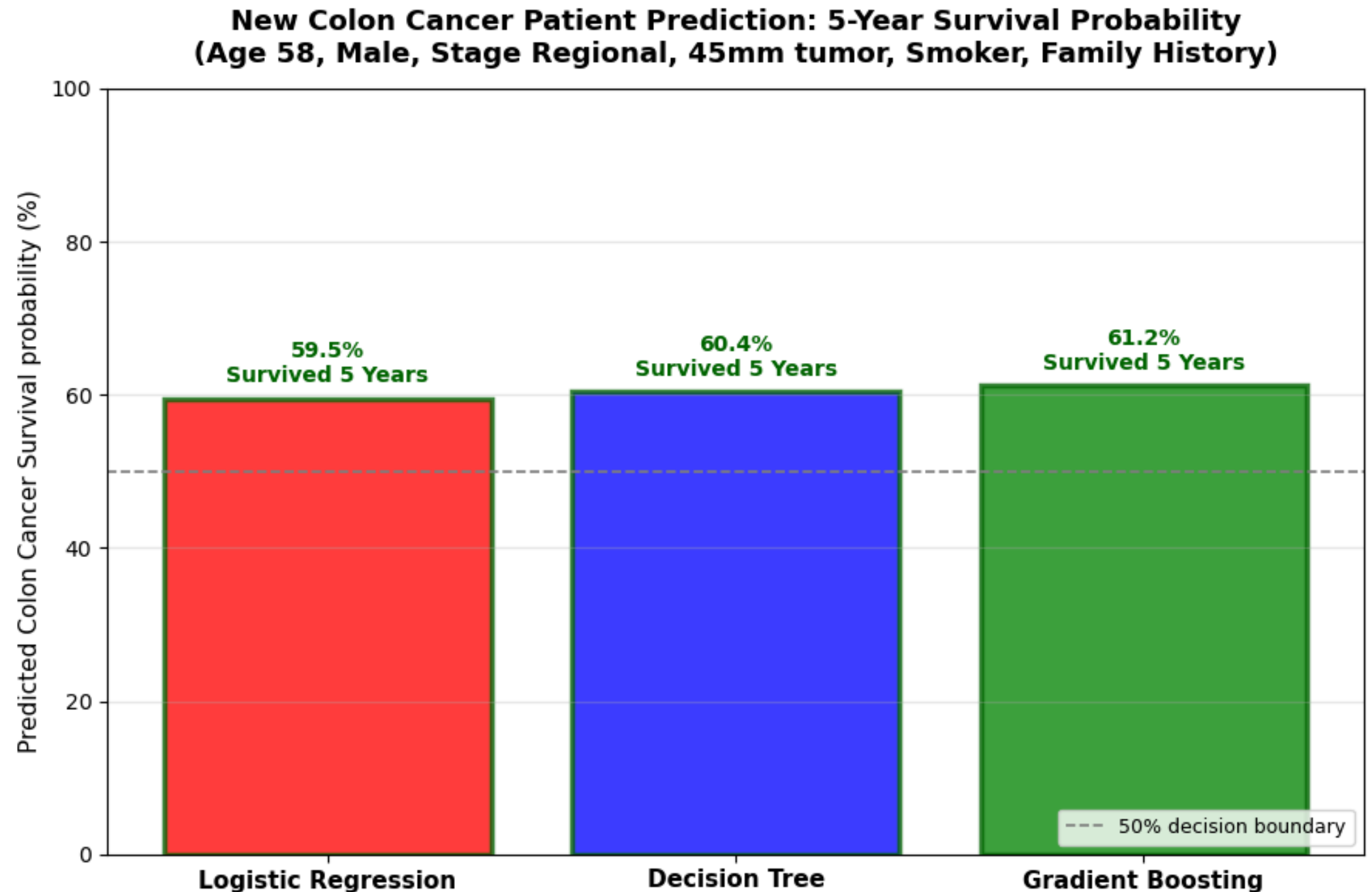
ModelResults: Colon Cancer Mortality Rate by Treatment Type

- Mortality rates are remarkably uniform across all stage × treatment combinations, hovering tightly around the overall rate of 40.1% (range: 39.38%–40.81%), suggesting treatment type has little differential impact on outcomes within each stage.
- The lowest mortality appears with Combination therapy for Regional (39.38%) and Metastatic (39.46%) stages, while the highest is Combination therapy for Localized cancer (40.81%), though these differences are small and likely not clinically meaningful.



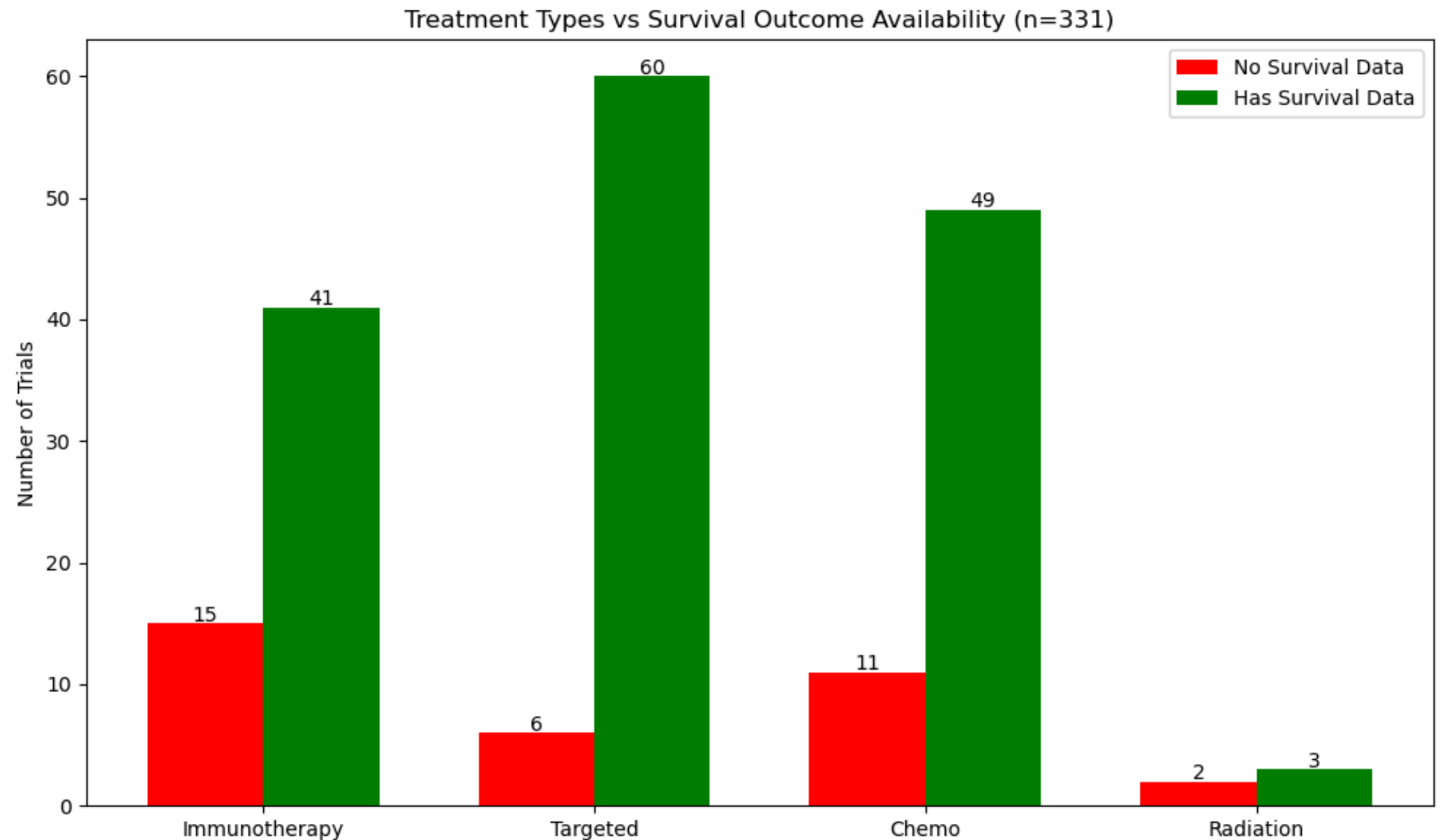
Model Results: New Colon Cancer Patient Survival Probability

- All three models agree the patient is predicted to survive 5 years, with survival probabilities clustered tightly between 59.5% (Logistic Regression) and 61.2% (Gradient Boosting). A spread of less than 2 percentage points, suggesting the signal in this patient's profile is consistent across model types.
- However, every prediction sits only ~10 points above the 50% decision boundary, meaning these are low-confidence "survives" rather than strong ones & clinically, the patient's risk factors (Stage Regional, 45mm tumor, smoker, family history) are pulling the predictions down from what would otherwise be a healthier baseline.



Model Results: Colon Cancer Clinical Trials

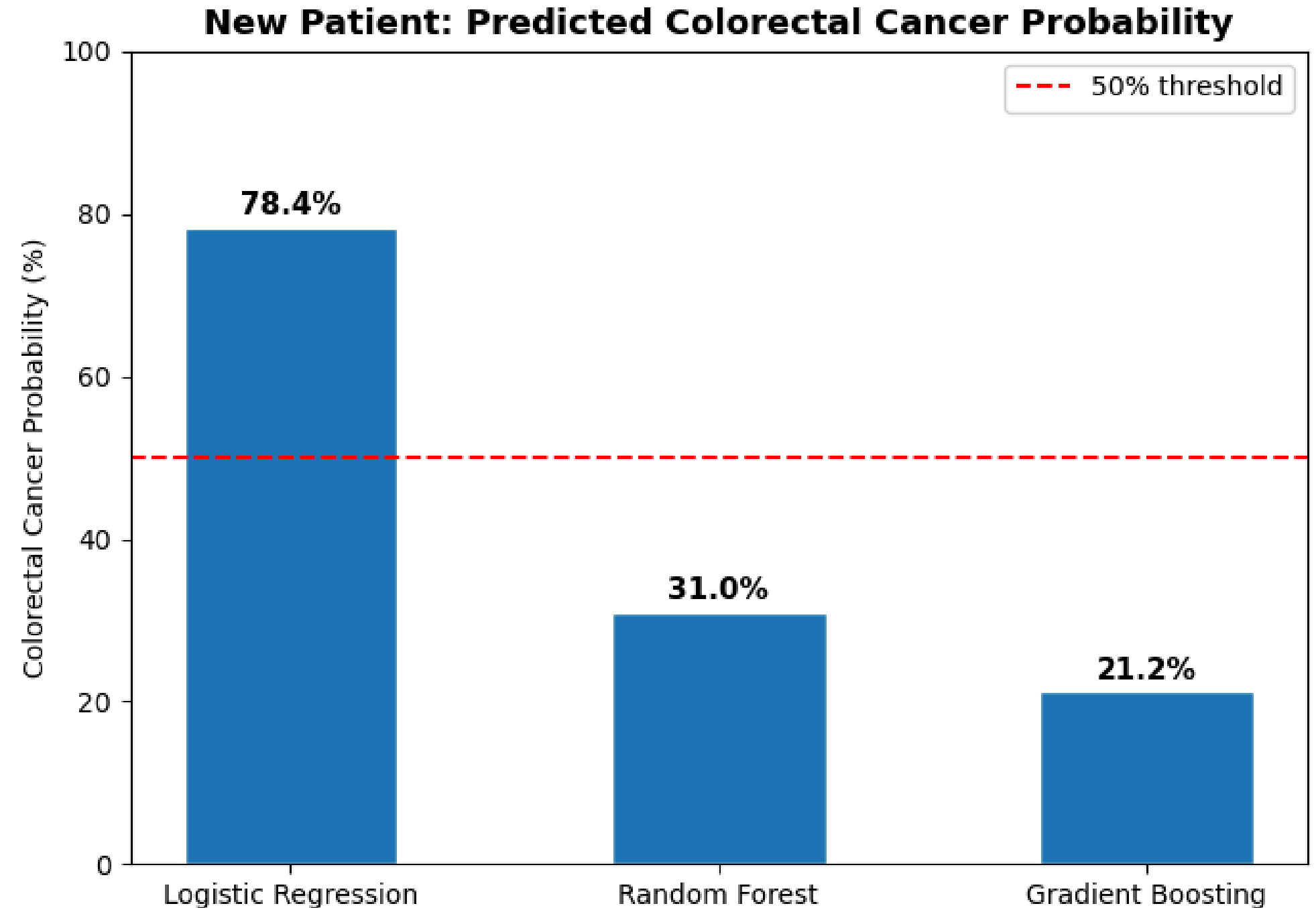
- Across 331 colon cancer trials, the majority report survival data, with Targeted therapy (60 of 66, 91%) and Chemo (49 of 60, 82%) showing the highest reporting rates, while Immunotherapy has the largest absolute gap (15 trials missing survival data out of 56).
- Radiation trials are notably underrepresented overall (only 5 trials), and uniquely have nearly as many missing as reporting survival outcomes (2 vs. 3).
- The chart is limited to showing trial counts by treatment type and whether survival data exists, it does not reveal any actual survival outcomes, effect sizes, trial quality, patient demographics, or other contextual details that would be needed to draw clinical conclusions.



Model Results: New Patient – Colon Cancer Probability

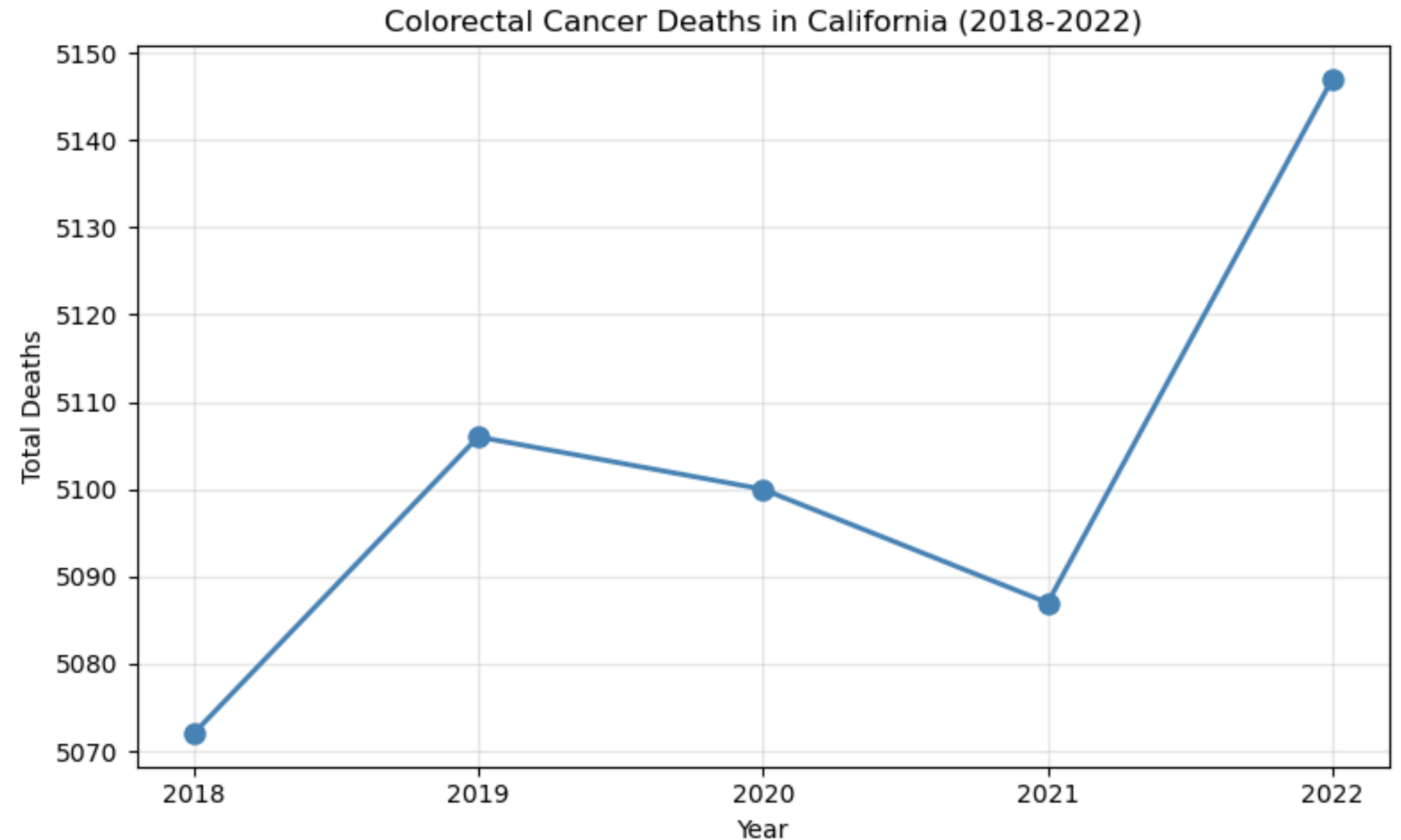
Based on data from `crc_dataset.csv`:

- Logistic Regression predicts a 78.4% probability of colorectal cancer, well above the 50% threshold, classifying the patient as high-risk.
- Random Forest (31.0%) and Gradient Boosting (21.2%) both fall below the 50% threshold, classifying the patient as low-risk.
- The models disagree significantly, suggesting the patient's features sit in a borderline region where the linear Logistic Regression model captures different patterns than the non-linear ensemble methods.

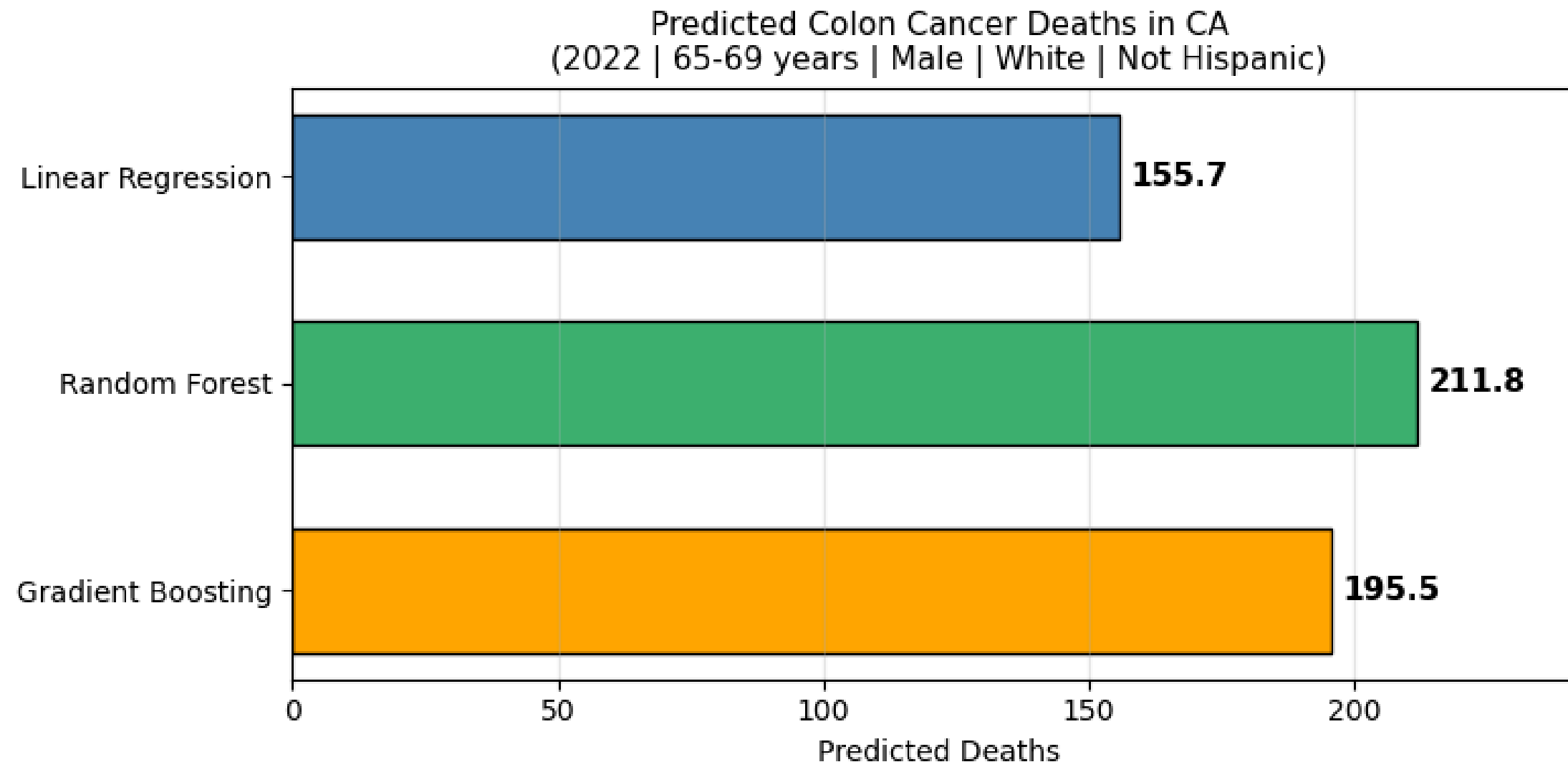


Model Results: Colon Cancer Deaths in CA (2018–2022)

- Accessed the CDC WONDER online database and queried the Underlying Cause of Death dataset, filtering by malignant neoplasms of the colon, rectosigmoid junction, rectum, and anus, with California selected as the state and the years 2018–2022 specified in the time period parameter.
- Grouped the results by year to extract annual death counts, then exported the data and visualized the five-year trend in a line chart showing total colorectal cancer deaths in California ranging from 5,072 in 2018 to 5,147 in 2022.



Model Results: New Patient Prediction in CA



- A new patient prediction was generated for a 65–69 year old White, Non-Hispanic male in California (2022), with the three models producing notably different estimates: Linear Regression predicted 155.7 deaths, Gradient Boosting 195.5, and Random Forest 211.8, a spread of roughly 56 deaths between the lowest and highest estimate.
- The tree-based models (Random Forest and Gradient Boosting) predicted higher mortality than Linear Regression, suggesting non-linear interactions between age, sex, and race that the linear model fails to capture for this high-risk demographic group.

Project Challenges

- API endpoints returned irrelevant data, requiring a switch to an alternative API source.
- HIPAA regulations limit access to individual patient data, restricting datasets to aggregated statistics.
- Medical data is fragmented, inconsistently formatted, and often contains suppressed values.
- Large CSV files demanded careful memory management and preprocessing before analysis.
- Class imbalance (survival cases far outnumbering deaths) hindered accurate mortality prediction.
- One of the datasets was synthetically generated, which introduced distributional skewness and unrealistic feature relationships that limited the model's ability to generalize to real-world clinical data.



The background features an abstract design with several thin green circles and solid green dots of varying sizes. A large circle is centered on the left, with a smaller one overlapping its top-left edge. Another large circle is partially visible on the right edge. Several dots are scattered throughout the composition, including a small one at the bottom left and a pair of dots at the bottom right. A small green triangle points towards the top right.

Questions ?

The background is white with several green circles of varying sizes. A large, thin green circle is centered behind the text. To its left is a smaller, solid green circle. Above the text, there is a small green dot and a green play button icon. Below the text, there are two more green dots. On the right side, there is a large, thin green circle and a small green dot. In the bottom left corner, there is a small green dot.

**Thank You for
Your
Attention !**